

Financial Goal Planner: An AI-Driven Goal-Based Investment Planning and Risk-Aware Recommendation System

A PROJECT REPORT
submitted by

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to
the APJ Abdul Kalam Technological University
in partial fulfillment of requirements for the award of degree
of
Bachelor of Technology
in
Computer Science and Engineering



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CAPE COLLEGE OF ENGINEERING
ALAPPUZHA
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DECLARATION

We, the undersigned, hereby declare that the project report entitled **Financial Goal Planner: An AI-Driven Goal-Based Investment Planning and Risk-Aware Recommendation System**, submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala, is a record of the work carried out by us under the guidance of Mr. Siju Koshy.

The report has been prepared based on the work completed during Phase 1 of the project. Wherever ideas or words of others have been used, the corresponding sources have been acknowledged and cited.

We further declare that the work presented in this report has not been submitted previously for the award of any degree, diploma, or similar title of any other University or institution.

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CERTIFICATE

This is to certify that the report entitled **Financial Goal Planner: An AI-Driven Goal-Based Investment Planning and Risk-Aware Recommendation System** submitted by **Amarthya Regimon Joseph** (PRP23CS008), **Aswin P** (PRP23CS023), **Athira Raj** (PRP23CS025) & **Haleema S** (PRP23CS041) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Individuals often find it difficult to plan investments for goals such as retirement, education, or vehicle purchase because the required savings, investment period, and acceptable risk vary with the goal timeline. Existing goal-based planning tools may use fixed risk categories and may not adapt recommendations to the time remaining for a goal. This project proposes a mobile application that provides personalized savings and investment planning across mutual funds, equity, and debt instruments based on a user’s financial goal, time horizon, and risk profile. The recommendation engine is inspired by the base paper, “Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards”, and adapts its regret-based and future-looking reward idea to goal-based, horizon-sensitive asset allocation.

The proposed system also includes inflation-adjusted goal calculation, SIP-related financial calculation, a Safety/Feasibility Calculator, risk profiling, dynamic allocation, and a mobile dashboard for tracking goal progress and recommended portfolio adjustments. Phase 1 establishes the literature foundation, research gap, problem statement, objectives, proposed architecture, and development plan for the system.

ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
DJI	Dow Jones Industrial Average
DRL	Deep Reinforcement Learning
GAF	Gramian Angular Field
LLM	Large Language Model
LSTM	Long Short-Term Memory
ML	Machine Learning
PPO	Proximal Policy Optimization
PSR	Probabilistic Sharpe Ratio
RL	Reinforcement Learning
RMS	Root Mean Square
SIP	Systematic Investment Plan
UI	User Interface

NOTATION

P	Principal amount
R	Expected rate of return
n	Investment period
FV	Future value
PV	Present value
SIP	Systematic Investment Plan
R_p	Portfolio return
σ	Portfolio risk/standard deviation

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Chapter 1

INTRODUCTION

1.1 Background

Financial planning is an important part of achieving long-term financial goals such as higher education, retirement, vehicle purchase, or other planned expenses. However, individuals may find it difficult to determine how much they need to save, how long they need to invest, and what level of investment risk is suitable for a particular goal.

The financial requirement of a goal depends on several factors, including the target amount, investment horizon, expected returns, inflation, and the amount that an individual can regularly save. Therefore, a financial plan that is suitable for one goal may not be appropriate for another goal with a different time horizon or financial requirement.

Traditional goal-based investment planning generally uses predefined investment categories or fixed allocation strategies. Such approaches may not adequately consider changes in the user's financial situation, the remaining time to achieve the goal, or the need to gradually modify the investment allocation as the goal approaches.

Artificial Intelligence (AI), Machine Learning (ML), and Deep Reinforcement Learning (DRL) provide opportunities for developing adaptive financial planning and recommendation systems. Reinforcement Learning can learn suitable actions based on the current state and the expected future outcomes. This makes it relevant to investment allocation problems where decisions may need to be adjusted over time.

The proposed project applies these concepts to develop a goal-based investment planning system that combines financial calculation, feasibility assessment, risk profiling, and adaptive investment recommendation.

1.2 Project Overview

The proposed project is a mobile-first Financial Goal Planner that helps users plan investments according to their specific financial goals. The system accepts goal-related and financial information from the user and uses this information to determine the financial requirements of the goal.

The system considers factors such as the target amount, time horizon, available savings, expected returns, and risk profile. Based on these inputs, the system is designed to calculate the required savings or periodic investment and assess whether the selected goal is financially feasible.

The project also includes a Risk Profiling module that considers the user's risk tolerance and risk capacity. The resulting risk profile is used together with the goal horizon to support the generation of personalized investment recommendations.

The Recommendation Engine is inspired by the base paper, *Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards*. The base paper uses Deep Reinforcement Learning with a regret-based and future-looking reward formulation for portfolio enhancement.

In the proposed project, these concepts are adapted to a goal-oriented investment planning context. Instead of considering portfolio performance alone, the system is designed around the user's financial goal, investment horizon, and risk profile.

The system also includes Dynamic Allocation. The recommended asset allocation can be adjusted according to the remaining time to the goal. As the goal approaches, the system can consider a more conservative allocation according to the user's risk profile and goal requirements.

The complete system is developed as a group project with the following major technical contribution areas:

- Mobile Application and User Interface.
- Financial Calculation and Safety/Feasibility Calculator.
- Risk Profiling.
- Recommendation Engine and Dynamic Allocation.

These modules are interconnected to provide an overall workflow from goal creation and financial assessment to risk profiling and investment recommendation.

1.3 Motivation

Planning investments for a specific financial goal involves more than selecting an investment product. The user must first understand the amount required for the goal, the time available to achieve it, the amount that can be invested periodically, and the level of risk that can be accepted.

A major motivation for the proposed system is to bring these activities together within a single application. Instead of treating financial calculation, risk assessment,

and investment recommendation as separate activities, the proposed system integrates them into one goal-based workflow.

The time horizon of a financial goal is particularly important in investment planning. A user with a long-term goal may have more time to accommodate changes in investment value, whereas a user approaching the goal deadline has less time to recover from unfavourable market movements. Therefore, the investment allocation should be considered in relation to the remaining goal horizon.

Another motivation is the application of AI and Deep Reinforcement Learning concepts to personalized financial recommendation. The base paper demonstrates the use of regret-based and future-looking reward concepts for portfolio enhancement. The proposed project explores how these concepts can be adapted to a goal-based setting, where the recommendation is influenced not only by portfolio performance but also by the user's goal and remaining investment period.

Thus, the project aims to develop a system that combines financial planning, feasibility assessment, risk profiling, and adaptive investment recommendation in a single mobile-first application.

1.4 Scope of Phase 1

Phase 1 establishes the foundation for the development of the proposed Financial Goal Planner. The main focus of this phase is understanding the problem, studying existing research, identifying the research gap, and planning the proposed system.

The first activity of Phase 1 was the detailed study of the base paper, *Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards*. The study helped in understanding the use of Deep Reinforcement Learning, regret-based rewards, and future-looking rewards in portfolio enhancement.

A literature review was then carried out to study existing work related to AI-based financial planning, robo-advisors, investor profiling, reinforcement learning for portfolio optimization, risk-aware investment, and goal-based financial planning. The review was used to identify limitations in existing approaches and to formulate the research gap for the proposed project.

Based on the literature study, the project problem statement and objectives were defined. The system requirements and major modules were also identified. The proposed architecture was planned to connect the mobile application with the financial calculation, feasibility, risk profiling, recommendation, and dynamic allocation modules.

The financial calculation component was planned to determine the financial

requirements associated with a goal, including the required future amount and periodic investment. The Safety and Feasibility Calculator was planned to assess whether the selected goal is practically achievable based on the available financial information.

The Risk Profiling module was planned to assess the user's risk tolerance and risk capacity. The Recommendation Engine was planned to use the goal information, time horizon, and risk profile to generate personalized investment recommendations.

The Dynamic Allocation component was planned to modify the asset allocation according to the remaining goal horizon. The overall mobile application and dashboard were also planned to present the goal information, financial calculations, risk profile, recommendations, and progress in a user-friendly manner.

Therefore, Phase 1 mainly focuses on research, analysis, system planning, and module-level design. The implementation and detailed evaluation of the proposed system are considered as subsequent phases of the project.

Chapter 2

LITERATURE REVIEW

The literature review was conducted to understand the existing research related to artificial intelligence in financial planning, robo-advisory, investor profiling, reinforcement learning, portfolio optimization, risk-aware investment, and goal-based financial planning. The reviewed studies provide the theoretical and technical foundation for the proposed Financial Goal Planner.

The literature mainly focuses on automated investment decisions, portfolio optimization, risk management, financial forecasting, investor behaviour, and AI-based advisory systems. Reinforcement learning and deep reinforcement learning have received particular attention because financial decisions are sequential in nature and the investment strategy may need to change according to market conditions.

The reviewed studies also indicate that conventional investment strategies may not be sufficient for all users because investment decisions depend on factors such as financial capacity, risk tolerance, investment horizon, expected returns, and changing market conditions. Therefore, the proposed project combines financial goal calculation, feasibility assessment, risk profiling, AI-based recommendation, and dynamic allocation into a unified goal-oriented system.

2.1 Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards

Daniil Karzanov, Ruben Garzón, Mikhail Terekhov, Caglar Gulcehre, Thomas Raffinot, and Marcin Detyniecki proposed an agent-based approach for improving an existing portfolio strategy using Deep Reinforcement Learning (DRL). The paper uses Proximal Policy Optimization (PPO) for dynamic portfolio rebalancing and considers the conventional 60/40 portfolio as the strategy to be enhanced.

A major contribution of the study is the use of a regret-based Sharpe reward function. Instead of simply rewarding the agent for obtaining a high return, the reward considers the difference between the performance of the selected allocation and an

optimal allocation. This allows the agent to learn from the opportunity that was missed by selecting a particular portfolio allocation.

The paper also introduces a future-looking reward function. Future return information is used during training to make the reward more representative of the future consequences of an allocation decision. However, future information is not used during inference, thereby avoiding information leakage during testing.

Another important component is the transaction cost scheduler. Transaction costs are gradually introduced during training so that the agent can learn the investment problem without being dominated by transaction-cost penalties at the beginning of training.

The study also uses synthetic financial data generated through a circular block bootstrap approach. This is intended to increase the diversity of training data and improve generalization.

The main evaluation measures considered in the paper are return and maximum drawdown. The paper therefore combines return optimization with risk considerations instead of focusing only on return.

The proposed project is directly inspired by this work. The Recommendation Engine uses the concept of regret-aware decision making, while Dynamic Allocation adapts the recommended allocation according to the remaining investment horizon. The proposed system extends the idea from general portfolio enhancement to goal-oriented financial planning.

2.2 AI-driven Robo-Advisors on Personal Financial Management and Investment Strategies

Dhashanamoorthi et al. studied the application of Artificial Intelligence in robo-advisory and personal financial management. Robo-advisors are automated systems that assist users in making investment-related decisions by processing financial and user information.

The study highlights the role of AI in automating financial decision-making and portfolio management. Instead of depending entirely on manual financial advice, an AI-based system can analyse user-related information and provide personalized recommendations.

One important aspect of robo-advisory is personalization. Different users have different financial conditions, investment objectives, and risk preferences. Therefore, a recommendation that is suitable for one user may not be suitable for another user.

The paper is relevant to the proposed project because the Financial Goal Planner also aims to provide personalized investment guidance. The proposed system collects information about the user's financial goal, target amount, investment horizon, savings capacity, and risk profile before generating a recommendation.

Thus, this study provides support for the use of AI-based personalized financial advisory as a component of the proposed system.

2.3 AI Revolution in Wealth Management: Applications and Horizons

Preetham Sunilkumar discussed the growing role of Artificial Intelligence in wealth management. The study focuses on how AI technologies can be applied to financial decision-making and wealth-management activities.

AI can process large amounts of financial information and can be used to support decision-making processes that may otherwise require significant manual effort. The application of AI can therefore improve automation and personalization in financial services.

The paper is important for the proposed project because the Financial Goal Planner is also designed as an AI-assisted financial planning system. The project uses automated calculations, risk assessment, recommendation generation, and dynamic allocation to support the user.

However, a general wealth-management system may not specifically focus on individual financial goals and deadlines. The proposed system therefore focuses on goal-oriented investment planning where the recommendation is connected to a specific financial objective and its time horizon.

2.4 A Multifaceted Approach to Stock Market Trading Using Reinforcement Learning

Ansari et al. presented a reinforcement learning based approach for stock market trading. The study investigates the use of multiple financial indicators and reinforcement learning techniques for making trading decisions.

The work demonstrates the relevance of reinforcement learning to financial decision-making. In reinforcement learning, an agent observes the state of the environment, performs an action, receives a reward, and learns a strategy that improves future decisions.

The paper is significant because financial markets are sequential and dynamic environments. An investment decision at one point in time can influence the portfolio state and future decisions.

The study is related to the proposed project through its use of reinforcement learning for financial decision-making. However, the objective of stock trading differs from the objective of the proposed Financial Goal Planner.

The proposed project does not focus primarily on short-term stock trading. Instead, it focuses on long-term financial goals and goal-based asset allocation. The reinforcement learning concept is therefore adapted toward allocation and recommendation rather than direct speculative trading.

2.5 A Novel RMS-Driven Deep Reinforcement Learning for Optimized Portfolio Management in Stock Trading

Sattar et al. investigated a Deep Reinforcement Learning approach for portfolio management. The work is relevant to the problem of automatically selecting and managing portfolio allocations in a changing financial environment.

The use of a risk-related performance measure is particularly important in portfolio management because a high return alone does not necessarily indicate a suitable investment strategy. Portfolio risk must also be considered when evaluating investment performance.

This study supports the idea that a portfolio management system should evaluate both return and risk. This principle is incorporated into the proposed Financial Goal Planner through risk profiling and risk-aware recommendation.

The main difference is that the proposed project combines portfolio allocation with financial-goal information. The recommended allocation is expected to depend not only on market information but also on the user's risk profile and remaining goal horizon.

2.6 Complex Forecasting and Investment Strategy Optimization via Chain-of-Thought of Large Language Models

Yin and Guo investigated the use of Large Language Models (LLMs) for financial forecasting and investment strategy optimization. The study represents the emerging use of generative AI and language models in financial applications.

LLMs can process and organize complex information and may provide additional support for financial analysis and decision-making. This introduces a different approach from conventional numerical financial models.

The paper is relevant to the proposed project because AI-based financial systems increasingly require mechanisms for converting complex financial information into understandable recommendations.

However, an LLM-based approach alone does not replace numerical financial calculations, risk assessment, or portfolio optimization. The proposed system therefore separates financial calculations, risk profiling, and recommendation generation into dedicated modules.

This provides a more structured architecture in which AI-based recommendation is supported by quantitative financial calculations.

2.7 Detecting and Adapting to Crisis Pattern with Context Based Deep Reinforcement Learning

Benhamou et al. studied the use of context-based Deep Reinforcement Learning for detecting and adapting to crisis patterns.

Financial markets can behave differently during normal market conditions and during crisis periods. Therefore, an investment strategy that performs well under normal conditions may not remain appropriate during periods of increased market uncertainty.

The study is relevant to the proposed project because dynamic investment allocation should consider changes in the financial environment. A goal-based portfolio may need to become more conservative when the financial environment becomes unfavourable.

The proposed system uses the remaining goal horizon and risk profile as important factors for allocation decisions. Crisis-aware techniques can be considered as a future

extension of the proposed Recommendation Engine and Dynamic Allocation modules.

2.8 Effectiveness of SIP Investment Among Retail Investors in Jharkhand: A Survey-Based Study

Samanta et al. studied the effectiveness and use of Systematic Investment Plans (SIPs) among retail investors.

SIP is an important investment mechanism for individuals who invest a fixed amount periodically rather than investing a large amount at once. It is particularly relevant to goal-based financial planning because a user can plan periodic investments to reach a future target.

The study provides background for considering periodic investment as an important component of personal financial planning.

The proposed Financial Goal Planner includes the calculation of required periodic investment or SIP contribution for achieving a future financial goal. Therefore, this literature supports the financial calculation component of the proposed system.

The proposed project further extends this concept by connecting the required SIP contribution with risk profile, feasibility, and recommended asset allocation.

2.9 Exploring the Potential of AI in Personal Financial Planning and Advisory Services

Ganesan, Vasantha, and Raja examined the potential applications of Artificial Intelligence in personal financial planning and advisory services.

Personal financial planning involves several activities including understanding financial conditions, setting objectives, evaluating investment options, and monitoring financial progress. AI can help automate parts of this process and provide personalized assistance based on user-specific information.

The study is relevant to the proposed project because the Financial Goal Planner attempts to combine these activities into one system. The proposed architecture includes goal management, financial calculation, feasibility assessment, risk profiling, recommendation, and progress monitoring.

The main contribution of the proposed project is the integration of these components with goal-oriented dynamic allocation.

2.10 Forecasting Inflation Using Disaggregates and Machine Learning

Medeiros and Boaretto studied the application of Machine Learning for inflation forecasting.

Inflation is important in financial planning because the purchasing power of money changes over time. A financial goal defined today may require a larger amount in the future because of inflation.

Therefore, financial planning systems should consider the future value of a goal rather than simply using its current cost.

The proposed Financial Goal Planner considers inflation while estimating the future goal amount. The inflation forecasting literature provides motivation for incorporating inflation into goal-based financial calculations.

The proposed project does not necessarily require a dedicated inflation forecasting model in the initial phase. Instead, the system can use an assumed or externally supplied inflation rate. More advanced inflation prediction can be considered as a future extension.

2.11 From Time Series to Images: A GAF-ResNet-LSTM Framework for Asset Selection and Portfolio Optimization

Solanki et al. proposed a framework that converts financial time-series information into image-like representations using Gramian Angular Fields (GAF) and combines deep learning techniques for asset selection and portfolio optimization.

Financial time series contain temporal patterns that may be difficult for conventional models to capture directly. Transforming time-series information into alternative representations can allow deep learning models to identify complex patterns.

The combination of GAF, ResNet, and LSTM demonstrates how different deep learning architectures can be combined for financial prediction and portfolio selection.

This work is relevant to the proposed project because the Recommendation Engine may use historical market information as part of future implementations.

However, the proposed project focuses more strongly on user goals, risk profile, and investment horizon rather than only predicting asset behaviour. Therefore, such deep learning forecasting techniques can be incorporated later as additional inputs to

the recommendation process.

2.12 Hybrid LSTM and PPO Networks for Dynamic Portfolio Optimization

Yugopuspito et al. proposed a hybrid approach combining Long Short-Term Memory (LSTM) networks with Proximal Policy Optimization (PPO) for dynamic portfolio optimization.

LSTM networks are suitable for sequential data because they can retain information from previous observations. PPO, on the other hand, is a reinforcement learning algorithm that can learn decision-making policies.

Combining forecasting and reinforcement learning can provide a framework where predicted market information is used by an RL agent to determine portfolio actions.

This work is highly relevant to the proposed project because PPO is also connected to the base paper and the planned recommendation engine.

The proposed project can use a similar separation of prediction and decision-making in future versions. However, the first phase focuses on the overall architecture and goal-based recommendation rather than implementing a complex hybrid forecasting model.

2.13 Machine Learning in Budget Forecasting for Corporate Finance: A Conceptual Model for Improving Financial Planning

Olamijuwon and Zouo investigated the use of Machine Learning in budget forecasting for financial planning.

Forecasting future financial requirements is an important part of financial planning. Machine Learning can assist in analysing historical information and estimating future requirements.

Although the paper focuses on corporate financial planning, the underlying idea is relevant to personal financial planning as well. A personal financial planning system also needs to estimate future requirements based on current information.

The proposed Financial Goal Planner applies this concept at the individual level. It estimates the future amount required for a financial goal and determines the required investment contribution.

Thus, the study provides supporting background for the financial calculation and feasibility modules.

2.14 Parallel and Multi-Stage Knowledge Graph Retrieval for Behaviorally Aligned Financial Asset Recommendations

Spadea and Seneviratne investigated knowledge-graph-based retrieval for financial asset recommendations.

Financial recommendation systems may need to consider several relationships among users, financial products, asset characteristics, and behavioural preferences.

Knowledge graphs provide a structured way of representing such relationships and retrieving relevant information.

This work is relevant to the proposed Recommendation Engine because personalized recommendation requires more than simply ranking financial assets by expected return.

The proposed project currently uses goal characteristics and risk profile as major inputs to recommendation. Knowledge-graph-based retrieval can be considered as a future extension for representing more detailed relationships between user preferences and financial assets.

2.15 Risk-Aware Deep Reinforcement Learning for Dynamic Portfolio Optimization

Lwele, Sabuni Emmanuel, and Sitali Gabriel investigated risk-aware Deep Reinforcement Learning for dynamic portfolio optimization.

Traditional reinforcement learning objectives may concentrate strongly on maximizing rewards. In financial applications, however, risk must also be considered because large losses can be harmful even when the average return is high.

A risk-aware RL approach is therefore important for investment applications where the objective is not simply maximum return.

This study directly supports the risk-aware design of the proposed project. The Financial Goal Planner includes a separate Risk Profiling module so that the recommendation can reflect the user's risk characteristics.

The proposed Dynamic Allocation module further considers the remaining goal horizon, allowing the investment recommendation to become more conservative as the deadline approaches.

2.16 Robo Advising and Investor Profiling

Gaspar and Oliveira studied robo-advisory and investor profiling.

Investor profiling is an important part of financial advisory because investors differ in their tolerance and capacity for financial risk.

An automated investment advisor should therefore collect relevant information about the investor before generating an investment recommendation.

This study directly supports the Risk Profiling module of the proposed system.

The proposed project considers risk profiling as an independent module. The output of the module is passed to the Recommendation Engine, where it is combined with goal information and investment horizon.

This separation helps ensure that recommendations are not generated without considering the user's financial characteristics.

2.17 Robo-Advisors in Financial Services: Redefining Wealth Management in the Age of Artificial Intelligence

Shetty, Singh, and Verma discussed the changing role of robo-advisors in financial services and wealth management.

Robo-advisors provide automated investment-related services and reduce the amount of manual intervention required in financial decision-making.

The increasing use of AI in wealth management indicates a shift towards automated and personalized financial services.

The paper is relevant to the proposed project because the Financial Goal Planner can be viewed as a goal-oriented robo-advisory application.

However, the proposed project has a narrower and more specific objective. Instead of general wealth management, it focuses on individual financial goals and determines how the recommended allocation should change according to the remaining goal period.

2.18 Scoring the Ethics of AI Robo-Advice: Why We Need Gateways and Ratings

Kofman discussed ethical considerations related to AI-based robo-advice.

Automated financial recommendations can influence important financial decisions. Therefore, recommendation systems should be designed with appropriate safeguards and should not be treated as unquestionable financial decisions.

This literature is important for the proposed project because the system provides recommendations based on user information and financial calculations.

The proposed system should therefore present recommendations as decision-support information rather than as guaranteed financial outcomes. The feasibility calculation and risk profiling modules also provide additional checks before presenting an allocation.

Ethical and responsible AI considerations can be incorporated into future versions through explainable recommendations, transparency, and appropriate user warnings.

2.19 Sentiment-Aware Portfolio Optimization: CVaR-Based Diversification with Deep Reinforcement Learning

Lamukanyani and Mantshimuli studied sentiment-aware portfolio optimization using Deep Reinforcement Learning and Conditional Value-at-Risk (CVaR)-based diversification.

The study demonstrates the possibility of combining market sentiment information with portfolio optimization.

Sentiment represents an additional information source that can potentially help an investment system understand market conditions. CVaR provides a way of considering losses beyond ordinary measures of average risk.

The paper is relevant to the proposed system because it demonstrates how alternative risk information and market signals can be combined with reinforcement learning.

The initial version of the proposed Financial Goal Planner does not depend on sentiment analysis. However, sentiment information can be considered as a future input to the Recommendation Engine, while risk measures such as CVaR can be explored for more advanced risk-aware allocation.

2.20 Simul-RL Portfolio Framework: Black-Scholes-Merton and Reinforcement Learning for Asset Allocation

Ahn and Kang proposed a portfolio framework combining financial modelling concepts with reinforcement learning for asset allocation.

The work demonstrates the use of a simulated financial environment to study reinforcement learning based allocation decisions.

Simulation is particularly useful in financial research because real market data may not contain enough examples of every possible market condition. Simulated environments can provide additional training scenarios for an RL agent.

This is related to the base paper, which also uses synthetic data generation to improve the learning process.

The proposed project can benefit from similar simulation concepts when testing dynamic allocation behaviour under different investment horizons and market conditions.

2.21 The Impact of the Digital Economy on Investment Portfolio Management in the Stock Market

Pereguda and Biloshkurskyi discussed the impact of digital transformation on investment portfolio management.

The growth of digital technologies has changed how financial information is collected, analysed, and used for investment decisions.

Digital financial systems can provide users with faster access to financial information and automated portfolio-management services.

The study provides broader motivation for developing a digital financial planning application.

The proposed Financial Goal Planner follows this direction by providing a mobile-friendly interface through which users can enter financial goals, view calculations, receive recommendations, and monitor progress.

2.22 The Role of Robo-Advisors in Behavioural Finance and Shaping Investment Decisions

Kulkarni, Patil, and Pramod studied the role of robo-advisors in behavioural finance and investment decisions.

Investor behaviour can influence financial decisions. Individuals may make decisions based on emotions, personal preferences, or short-term market reactions.

Automated advisory systems can provide structured recommendations based on predefined financial information and investment objectives.

This is relevant to the proposed project because the system uses a structured risk-profiling and recommendation process rather than depending entirely on spontaneous user investment decisions.

The proposed system can also be extended in the future to consider behavioural factors as additional inputs to the Risk Profiling module.

2.23 Uncertainty-Aware Reinforcement Learning for Portfolio Optimization

Enkhsaikhan and Jo investigated uncertainty-aware reinforcement learning for portfolio optimization.

Financial markets contain substantial uncertainty. A model may perform well under historical conditions but may behave differently when market conditions change.

Considering uncertainty is therefore important when reinforcement learning is applied to financial decision-making.

This study supports the need for cautious and risk-aware recommendations in the proposed Financial Goal Planner.

The proposed system addresses this at the planning level by considering user risk profile and investment horizon. More advanced uncertainty estimation can be integrated into the Recommendation Engine in future development phases.

2.24 AI-driven Goal Based Financial Planning System: A Framework for Contextual Feasibility Validation

Patra et al. proposed an AI-driven framework for goal-based financial planning with contextual feasibility validation.

Goal-based financial planning is directly related to the objective of the proposed project. Instead of recommending investments without considering a specific objective, goal-based planning starts with the user's target and determines the financial requirements for achieving it.

Feasibility validation is particularly important because a financial goal may not always be achievable using the user's available savings and investment horizon.

This study is therefore highly relevant to the proposed Safety and Feasibility Calculator.

The proposed project extends the goal-based planning concept by combining feasibility assessment with risk profiling, AI-based recommendation, and dynamic allocation.

The combination of these modules allows the system to move from simple goal calculation towards a complete goal-oriented investment planning workflow.

2.25 Comparative Summary of the Reviewed Literature

The reviewed literature demonstrates that AI and Machine Learning are increasingly being applied to financial planning, portfolio management, risk assessment, forecasting, and robo-advisory.

The literature can be broadly grouped into four major categories. The first category consists of reinforcement learning and deep reinforcement learning approaches for portfolio optimization. The second category focuses on robo-advisory and investor profiling. The third category addresses financial planning, forecasting, and goal-based investment. The fourth category considers additional factors such as sentiment, behavioural finance, uncertainty, and ethical considerations.

The major observations from the literature are summarized below.

Table 2.1: Summary of Reviewed Literature

Ref	Title	Methodology	Result	Advantages	Disadvantages
1	Regret-Optimized Portfolio Enhancement [BASE PAPER]	Regret-based portfolio optimization using risk-adjusted objectives and historical market data.	Produces portfolios with improved regret performance and better control of downside risk.	Improves portfolio robustness and risk control.	Requires reliable market estimates and can be computationally demanding.
2	AI-driven robo-advisors on personal financial management	AI-based robo-advisory framework using financial analysis, investor profiling and personalized recommendations.	Demonstrates the potential of AI to automate personalized financial-management decisions.	Automated and personalized advice with reduced advisory effort.	Dependent on data quality and has limited human interaction.
3	AI Revolution in Wealth Management	Analysis of AI and machine-learning applications in wealth-management and investment decisions.	Shows that AI can improve automation, financial analysis and decision-support capabilities.	Enables automation, scalability and faster financial analysis.	Data dependency, model complexity and explainability concerns.
4	Multifaceted Approach to Stock Market Trading Using RL	Reinforcement-learning framework using market information and trading strategies for automated stock trading.	RL-based strategies demonstrate the ability to adapt trading decisions to changing market conditions.	Adaptive decision making and automated trading.	Sensitive to market noise, reward design and training parameters.
5	RMS-Driven Deep RL for Optimized Portfolio Management	Deep reinforcement learning combined with risk-management techniques for dynamic portfolio optimization.	Demonstrates improved portfolio optimization by incorporating risk management into the learning process.	Combines return optimization with risk management.	High computational requirements and possible training instability.

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Table 2.1 – continued from previous page

Ref	Title	Methodology	Result	Advantages	Disadvantages
6	Complex forecasting via chain-of-thought of LLMs	Large language models with chain-of-thought reasoning for complex forecasting tasks.	Shows that structured LLM reasoning can improve performance on complex forecasting problems.	Improves reasoning capability for complex forecasting.	High computational cost and difficulty verifying generated reasoning.
7	Detecting and adapting to crisis pattern with context DRL	Context-aware deep reinforcement learning for detecting and adapting to financial crisis patterns.	The framework enables portfolio strategies to adapt when crisis-related market patterns are detected.	Adapts portfolio decisions during changing and crisis conditions.	Requires sufficient crisis-period data and complex training.
8	Effectiveness of SIP Investment Among Retail Investors	Empirical analysis of systematic investment plans using investor data and statistical analysis.	The study indicates the role of systematic investing in promoting disciplined investment behaviour among retail investors.	Highlights disciplined investing and investor behaviour.	Limited applicability to automated portfolio optimization.
9	Potential of AI in personal financial planning	AI-based personal financial planning using financial information and intelligent recommendation techniques.	Shows that AI can support personalized financial planning and automated recommendations.	Supports personalized planning and automated decision making.	Privacy, data-quality and explainability concerns.
10	Forecasting inflation using disaggregates and ML	Machine-learning models using disaggregated economic variables for inflation forecasting.	Machine-learning approaches demonstrate the ability to capture nonlinear relationships in inflation forecasting.	Captures complex economic relationships.	Sensitive to data quality, feature selection and model choice.

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Table 2.1 – continued from previous page

Ref	Title	Methodology	Result	Advantages	Disadvantages
11	GAF-ResNet-LSTM Framework for Asset Selection	Gramian Angular Field representation combined with ResNet and LSTM for financial asset selection.	The hybrid framework effectively captures spatial and temporal patterns for asset-selection decisions.	Captures both spatial and temporal financial patterns.	High model complexity and computational requirements.
12	Hybrid LSTM and PPO Networks for Portfolio Optimization	Hybrid architecture combining LSTM prediction with PPO reinforcement learning for portfolio optimization.	The hybrid model combines sequential market prediction with adaptive portfolio allocation.	Combines prediction with adaptive portfolio decisions.	Training complexity and sensitivity to hyperparameters.
13	ML in budget forecasting for corporate finance	Machine-learning models applied to corporate budget and financial forecasting.	ML-based forecasting can improve automated budget prediction and identify complex financial patterns.	Automates forecasting and identifies complex patterns.	Dependent on historical data quality and model assumptions.
14	Parallel and Multi-Stage Knowledge Graph Retrieval	Parallel and multi-stage knowledge-graph retrieval for structured information processing.	The approach improves information retrieval by combining multiple retrieval stages and contextual knowledge.	Improves retrieval efficiency and contextual information access.	Increased system complexity and computational overhead.
15	AI-Driven Goal Based Financial Planning System	AI-based financial planning using user goals, financial information and automated recommendations.	The system supports goal-oriented financial planning through personalized AI-based recommendations.	Provides personalized and goal-oriented financial planning.	Requires detailed user information and reliable financial data.
16	Risk-Aware Deep RL for Dynamic Portfolio Optimization	Risk-aware deep reinforcement learning for dynamic portfolio allocation.	The framework incorporates portfolio risk into the learning process to support risk-sensitive allocation decisions.	Explicitly incorporates investment risk.	Complex training and sensitivity to market conditions.

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Table 2.1 – continued from previous page

Ref	Title	Methodology	Result	Advantages	Disadvantages
17	Robo Advising and Investor Profiling	Robo-advisory framework using investor profiling and automated portfolio recommendations.	Shows how investor profiles can be incorporated into automated robo-advisory recommendations.	Provides scalable and personalized investment advice.	May provide less personalization than human advisory.
18	Robo-advisors in financial services	Review and analysis of AI-powered robo-advisory systems in financial services.	The review identifies automation, accessibility and scalability as major capabilities of robo-advisory systems.	Highlights automation, accessibility and scalability.	Trust, transparency, regulation and explainability concerns.
19	Scoring the Ethics of AI Robo-Advice	Ethical evaluation framework for assessing AI-based robo-advisory practices.	Provides a structured approach for identifying and evaluating ethical issues in AI-based robo-advice.	Provides a framework for ethical evaluation.	Ethical assessment may depend on subjective criteria.
20	Sentiment-Aware Portfolio Optimization (CVaR-Based)	Portfolio optimization combining market sentiment with CVaR-based downside-risk management.	The framework incorporates investor sentiment and downside risk into portfolio allocation decisions.	Combines sentiment analysis with downside-risk control.	Dependent on sentiment quality and accurate CVaR estimation.
21	Simul-RL Portfolio Framework: BSM and RL	Simulation framework combining Black-Scholes-Merton modelling with reinforcement learning.	Demonstrates how financial modelling and RL can be integrated for portfolio decision making.	Combines financial modelling with adaptive RL decisions.	Simulation assumptions may not fully represent real markets.
22	Impact of the Digital Economy on Investment Portfolio Mgmt	Analysis of digital-economy technologies and their impact on investment portfolio management.	Shows the influence of digital technologies on modern portfolio-management processes.	Highlights potential benefits of digitalization.	Primarily analytical with limited direct implementation.

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Table 2.1 – *continued from previous page*

Ref	Title	Methodology	Result	Advantages	Disadvantages
23	Role of robo-advisors in behavioural finance	Analysis of robo-advisory systems from a behavioural-finance perspective.	Shows the relationship between automated investment services and behavioural aspects of investors.	Connects automated investing with investor behaviour.	Behavioural factors are difficult to quantify and model.
24	Uncertainty-Aware RL for Portfolio Optimization	Reinforcement-learning framework incorporating uncertainty into portfolio optimization.	The framework accounts for uncertainty during portfolio decision making and allocation.	Accounts for uncertainty in portfolio decisions.	Higher computational complexity and difficult uncertainty estimation.

2.26 Overall Observation from Literature

The literature review shows that existing research has addressed many individual components required for intelligent financial decision-making. Deep Reinforcement Learning has been used for dynamic portfolio optimization, while robo-advisory research has focused on personalized investment services and investor profiling. Other studies have addressed financial forecasting, inflation, systematic investment, behavioural finance, uncertainty, sentiment, and ethical aspects of AI-based financial advice.

However, these research directions are often considered separately. A financial planning system intended for individual users requires multiple components to work together. In particular, the financial goal, target amount, time horizon, savings capacity, risk profile, and investment recommendation should be connected.

The literature therefore indicates the need for an integrated goal-oriented framework that combines financial calculation, feasibility assessment, risk profiling, personalized recommendation, and dynamic allocation.

The proposed Financial Goal Planner addresses this requirement by combining these components into a single system. The system begins with a user's financial goal and calculates the future financial requirement. It then evaluates feasibility, determines the user's risk profile, generates an investment recommendation, and adjusts the allocation according to the remaining goal horizon.

The Recommendation Engine is conceptually inspired by the regret-aware and future-looking reward approach of the base paper. The base paper specifically uses PPO, a regret-based Sharpe reward, future-looking information during training, synthetic data generation, and transaction-cost scheduling for dynamic portfolio rebalancing. These ideas provide the main technical foundation for the proposed recommendation and allocation approach.

The proposed project differs in its application objective. Rather than concentrating only on enhancing a general portfolio strategy, the proposed system is designed around an individual financial goal. Therefore, the investment recommendation is expected to be dependent on both financial-market considerations and user-specific goal characteristics.

Chapter 3

RESEARCH GAP AND PROBLEM STATEMENT

3.1 Research Gap

Robo-advisory and AI financial planning literature covers general AI-driven advisory concepts, investor profiling, ethics, and Indian SIP adoption.

However, the reviewed literature does not integrate a dynamic, reward-shaped recommendation engine tied specifically to a goal's remaining time horizon.

The proposed project focuses on combining:

1. Goal-based horizon-aware planning involving the target amount, deadline, and inflation-adjusted target.
2. A mobile-first, India-relevant recommendation engine spanning mutual funds, equity, and debt instruments.
3. Regret-based and future-looking Deep Reinforcement Learning reward concepts inspired by the base paper.

The project therefore aims to bridge goal-based financial planning with regret-based and future-looking DRL reward shaping in a single mobile recommendation system.

3.2 Problem Statement

Individuals planning for financial goals such as retirement, a child's education, or purchasing a vehicle face difficulty determining how much to save, for how long, and where to invest.

The appropriate level of investment risk depends heavily on the goal's remaining time horizon.

There is a need for a system that dynamically adjusts investment recommendations based on both the time remaining to a goal and prevailing market conditions, rather than relying on fixed, one-time risk classifications.

Chapter 4

OBJECTIVES AND PROPOSED SYSTEM

4.1 Objectives

The main objectives of the proposed project are:

1. To design a mobile application that allows users to define financial goals with a target amount and time horizon such as retirement, education, or vehicle purchase.
2. To develop a risk-profiling module that assesses both risk tolerance and risk capacity based on the user's goal horizon and financial situation.
3. To build a recommendation engine inspired by the base paper's regret-optimized and future-looking reward formulation that dynamically adjusts asset allocation across mutual funds, equity, and debt instruments as the goal's deadline approaches.
4. To provide an intuitive mobile dashboard for tracking goal progress, required SIP contributions, and recommended portfolio adjustments over time.

4.2 System Requirements

Table 4.1: Functional Requirements of the Proposed System

No.	Requirement	Description
1	Goal Creation	User can create a financial goal with target amount and time horizon.
2	Financial Calculation	System calculates required savings and periodic investment.
3	Risk Profiling	System evaluates user risk tolerance and risk capacity.
4	Feasibility Assessment	System checks whether the selected goal is financially achievable.
5	Investment Recommendation	System provides personalized investment allocation recommendations.
6	Dynamic Allocation	System adjusts allocation based on remaining goal horizon.
7	Progress Tracking	User can monitor goal progress through the mobile dashboard.

4.2.1 Functional Requirements

The proposed system shall provide the following functional capabilities:

- User profile management.
- Financial goal creation and management.
- Goal amount and deadline input.
- Inflation-adjusted goal calculation.
- SIP calculation.
- Safety and feasibility assessment.
- Risk profiling.
- Personalized investment recommendation.
- Dynamic asset allocation.
- Goal progress tracking.
- Mobile dashboard.

4.2.2 Non-Functional Requirements

The system should satisfy the following non-functional requirements:

- Usability.
- Reliability.
- Security.
- Maintainability.
- Scalability.
- Performance.
- Responsiveness.

4.2.3 Software Requirements

The planned software environment includes:

- Mobile application development framework.
- Backend and API services.
- Database system.
- Python-based machine-learning environment.
- Reinforcement Learning libraries.
- Development and testing tools.

4.3 Proposed System Architecture

The proposed system architecture illustrates the overall workflow of the Financial Goal Planner and the interaction between its major modules. The system collects the user's financial goal, target amount, investment horizon, savings capacity, and risk-related information through the mobile application.

The collected information is processed through the Goal Management, Financial Calculation, Safety and Feasibility, and Risk Profiling modules. These outputs are provided to the Recommendation Engine, which generates a personalized investment recommendation based on the user's goal, financial condition, risk profile, and investment horizon.

The Dynamic Allocation Module adjusts the recommended allocation according to the remaining goal horizon. Finally, the Dashboard presents the goal progress, financial requirements, feasibility status, risk profile, and recommended allocation to the user.

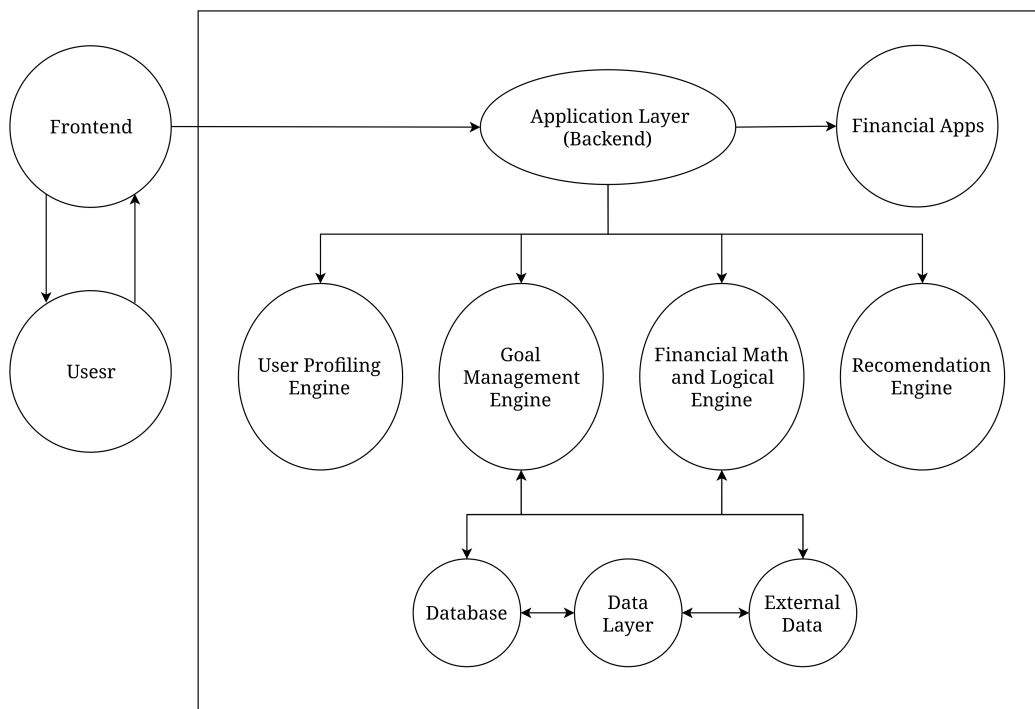


Figure 4.1: Proposed System Architecture

The proposed architecture is inspired by the base paper, “Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards”. The proposed system adapts the regret-aware and future-looking portfolio concepts to a goal-based financial planning environment by considering the financial goal, feasibility, risk profile, and remaining investment horizon.

4.4 Major Modules

The proposed system consists of the following major modules, which work together to provide goal-based and risk-aware investment planning.

4.4.1 Goal Management

The Goal Management module allows users to create and manage financial goals by specifying the goal type, target amount, and investment horizon. These details are used by the subsequent financial calculation and recommendation modules.

4.4.2 Financial Calculation Module

The Financial Calculation Module determines the financial requirements associated with the selected goal. It considers factors such as target amount, investment period, inflation, expected return, and required periodic contribution such as SIP.

4.4.3 Safety and Feasibility Calculator

The Safety and Feasibility Calculator evaluates whether the selected financial goal is practically achievable. It compares the required investment with the user's available savings or periodic contribution capacity.

4.4.4 Risk Profiling

The Risk Profiling module evaluates the user's risk characteristics using relevant financial and goal-related information. The resulting risk profile is used by the Recommendation Engine while generating the investment recommendation.

4.4.5 Recommendation Engine

The Recommendation Engine combines the goal details, financial calculations, feasibility result, risk profile, and investment horizon to generate a personalized investment recommendation. Its design is inspired by the regret-based and future-looking reward concepts of the base paper.

4.4.6 Dynamic Allocation and Dashboard

The Dynamic Allocation module adjusts the recommended asset allocation according to the remaining goal horizon and the user's risk profile. The Dashboard presents the goal progress, financial requirements, feasibility status, risk profile, and recommended allocation to the user.

4.4.7 Data Flow Diagram

The Data Flow Diagram (DFD) represents the flow of information between the user, the external data sources and the major modules of the proposed system. It shows how the user's financial and goal-related information is processed to generate the final investment recommendation. The system is presented at two levels: Level 0 (context diagram) and Level 1 (module-level decomposition).

Table 4.2: Major Modules of the Proposed System

No.	Module	Description
1	Goal Management	Defines the financial goal, target amount, and time horizon.
2	Financial Calculation	Calculates future goal amount, required savings, and SIP contribution.
3	Safety and Feasibility	Evaluates whether the selected financial goal is practically achievable.
4	Risk Profiling	Assesses the user's risk tolerance and risk capacity.
5	Recommendation Engine	Generates personalized investment recommendations based on goal, horizon, and risk profile.
6	Dynamic Allocation and Dashboard	Adjusts asset allocation and displays goal progress and recommendations.

Level 0 DFD (Context Diagram)

The Level 0 DFD shows the Financial Goal Planner as a single process and describes its interaction with external entities. The **User** provides goal details (target amount, time horizon) and risk-profile responses, and receives the inflation-adjusted goal amount, required SIP contribution, recommended asset allocation and portfolio adjustment alerts. The **Market Data Source** (mutual fund NAV data) supplies the historical and latest market data used for growth estimation and allocation.



Figure 4.2: Data Flow Diagram Level 0 (Context Diagram)

Level 1 DFD

The Level 1 DFD decomposes the system into its four major modules and shows the data exchanged between them and the central database:

- **Mobile Application & UI/UX:** Accepts goal details from the user, displays the dashboard and presents recommendations.
- **Risk Profiling & User Modelling:** Processes the questionnaire responses to assess risk tolerance and risk capacity, and outputs the user's risk class and financial profile.
- **Financial Forecasting & Computation:** Computes the inflation-adjusted goal amount, required SIP contribution, expected investment growth and goal feasibility using processed market data.

- **AI Recommendation & Portfolio Allocation:** Takes the risk profile, goal parameters and forecast results to generate the asset allocation across mutual funds, equity and debt instruments, and to adjust it as the remaining horizon shrinks.

User and goal data and generated recommendations are stored in and retrieved from the database, so that the dashboard can track goal progress over time.

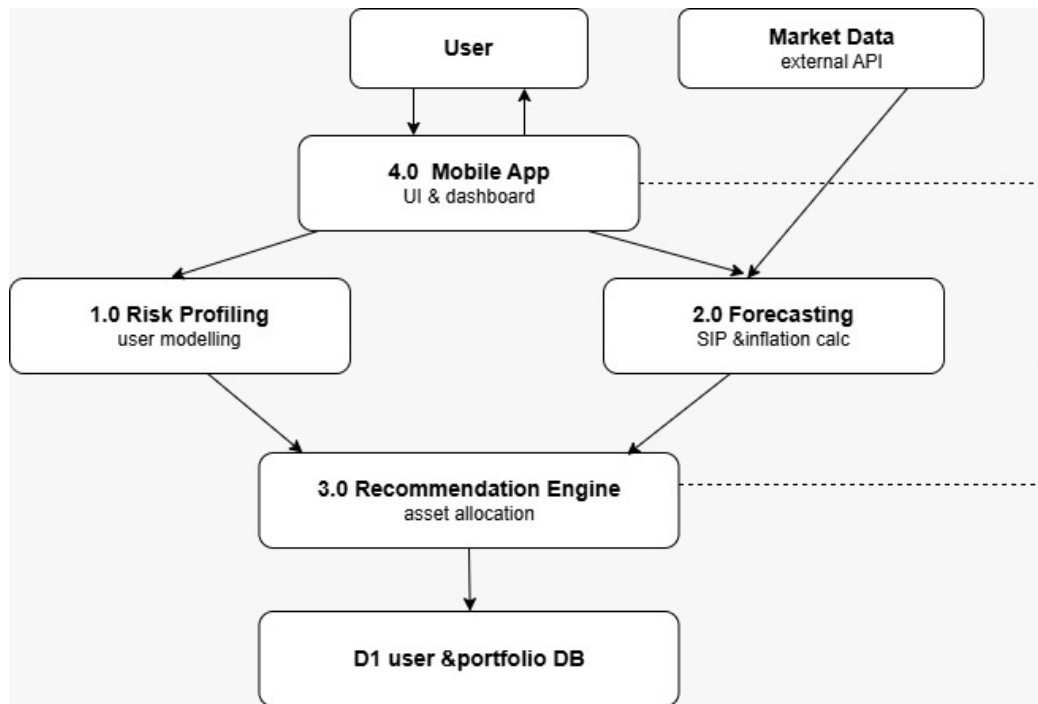


Figure 4.3: Data Flow Diagram Level 1

Chapter 5

PHASE 1 METHODOLOGY AND WORK COMPLETED

5.1 Phase 1 Methodology

Phase 1 was conducted as a collaborative group activity focused on understanding the problem domain, studying existing research, and designing the proposed Financial Goal Planner system. The group initially studied the base paper, “Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards”, to understand its approach to portfolio optimization, regret-based reward formulation, and future-looking rewards. The group examined how these concepts could be adapted from general portfolio optimization to a goal-based investment planning environment.

Following the study of the base paper, an extensive literature review was conducted to understand existing approaches related to Reinforcement Learning, Deep Reinforcement Learning, portfolio optimization, robo-advisory, personal financial planning, investor profiling, risk-aware investment, and Artificial Intelligence. The review helped the group understand the existing solutions, commonly used techniques, and limitations of current financial planning and investment recommendation systems.

Based on the findings from the literature review, the group identified the research gap addressed by the proposed system. The research gap was mainly focused on the need for a goal-oriented financial planning system that combines financial calculations, feasibility assessment, risk profiling, personalized investment recommendations, and dynamic asset allocation according to the user’s goal horizon.

The problem statement was then formulated to clearly define the limitations of existing approaches and the need for an integrated goal-based and risk-aware financial planning system.

The group subsequently defined the major objectives of the project. The objectives were formulated to cover financial goal management, financial calculations, safety and feasibility assessment, risk profiling, personalized investment recommendation, and dynamic asset allocation.

The system requirements were then identified based on the proposed objectives. Functional and non-functional requirements were considered along with the required software, hardware, and system-level components. The major modules of the proposed system were also identified and their responsibilities were defined.

The proposed system architecture was designed to represent the flow of information between the mobile application, financial calculation modules, safety and feasibility calculator, risk profiling, recommendation engine, dynamic allocation module, and dashboard. The architecture was designed to provide a clear understanding of how user inputs are processed to generate personalized investment recommendations.

Finally, the group prepared the planned development workflow for the subsequent phases of the project. The planned work includes detailed system design, mobile application development, financial calculation implementation, risk profiling, recommendation engine development, dynamic allocation, backend and database integration, model development and

evaluation, testing, and final documentation.

Thus, Phase 1 established the theoretical foundation, research direction, system requirements, architecture, major modules, and development plan required for the subsequent implementation phases.

Table 5.1: Major Inputs and Outputs of the System

Input	Output
Goal type	Selected financial goal
Target amount	Required future goal amount
Time horizon	Goal-based investment period
Financial situation	Risk capacity assessment
Risk tolerance	Risk profile
Savings capacity	Required SIP contribution and feasibility status
Risk profile and goal horizon	Recommended asset allocation

5.2 Literature Study

The literature study covered 24 related research works and the base paper.

The reviewed literature was analysed with respect to:

- Reinforcement Learning.
- Deep Reinforcement Learning.
- Portfolio optimization.
- Risk-aware investment.
- Robo-advisory.
- Investor profiling.
- Financial planning.
- Inflation forecasting.
- Behavioural finance.
- AI-based recommendations.
- Digital investment management.

The literature review helped the group identify the research gap and define the proposed system.

5.3 Financial Calculation and Feasibility Work

The Financial Calculation component focuses on estimating the financial requirements associated with achieving a selected goal.

The system considers:

- Current goal amount.
- Inflation rate.
- Future goal amount.
- Investment horizon.
- Expected investment return.
- Required monthly contribution.
- Available monthly savings.

The Safety and Feasibility Calculator uses these values to provide an initial assessment of whether the selected goal is financially feasible.

This module forms an important input to the later recommendation process.

5.4 Planned Recommendation Workflow

The planned workflow of the proposed system is:

1. User enters financial goal details.
2. System determines the goal horizon and target amount.
3. Inflation-adjusted goal value is calculated.
4. Required savings or SIP contribution is calculated.
5. Safety and Feasibility Calculator evaluates the goal.
6. Risk Profiling module evaluates the user's risk tolerance and risk capacity.
7. Recommendation Engine processes the goal horizon and risk information.
8. Initial asset allocation is generated.
9. Dynamic Allocation adjusts the recommendation as the goal deadline approaches.
10. Dashboard displays goal progress and recommendation details.

Table 5.2: Individual Contributions

Member	Assigned Contribution
Amarthya Regimon Joseph	Mobile Application and User Interface
Aswin P	Financial Calculation and Safety/Feasibility Calculator
Athira Raj	Risk Profiling
Haleema S	Recommendation Engine and Dynamic Allocation

5.5 Individual Contributions

The project is a collaborative group project consisting of four members. Each member has a specific technical contribution while all members jointly participate in the overall project development, literature study, documentation, and system planning.

- **Amarthya Regimon Joseph:** Responsible for the Mobile Application and User Interface development. The contribution includes planning the mobile-first interface, screen structure, navigation flow, user interaction, and presentation of financial goals, goal progress, feasibility status, and investment recommendations.
- **Aswin P:** Responsible for the Financial Calculation and Safety/Feasibility Calculator modules. The contribution includes planning the calculation of future goal amounts, inflation-adjusted target amounts, required savings, SIP contributions, expected returns, and preliminary feasibility assessment of the selected financial goal.
- **Athira Raj:** Responsible for the Risk Profiling module. The contribution includes planning the assessment of user risk tolerance and risk capacity using financial, personal, and goal-related information. The module is intended to provide a suitable risk profile for use by the recommendation system.
- **Haleema S:** Responsible for the Recommendation Engine and Dynamic Allocation modules. The contribution includes planning the recommendation mechanism inspired by the regret-optimized and future-looking reward formulation of the base paper. The dynamic allocation component considers the user's risk profile and remaining goal horizon to adjust the recommended asset allocation.

All four members jointly contributed to the literature review, research gap identification, problem statement, objectives, requirements analysis, proposed system architecture, Phase 1 methodology, documentation, and overall project planning.

5.6 Planned Development Schedule

The subsequent development phases are planned to progressively transform the Phase 1 design into a functional Financial Goal Planner. The planned activities include the following:

1. Detailed system design and database design.

2. Development of the mobile application and user interface.
3. Implementation of financial calculations, including future goal amount, inflation adjustment, required savings, and SIP contribution calculations.
4. Implementation of the Safety and Feasibility Calculator to assess the practical achievability of the selected financial goal.
5. Development of the Risk Profiling module for assessing user risk tolerance and risk capacity.
6. Development of the Recommendation Engine for generating goal-based and risk-aware investment recommendations.
7. Implementation of the Dynamic Allocation module to adjust asset allocation according to the remaining goal horizon.
8. Backend and database integration with the mobile application.
9. Model development, training, and evaluation for the planned AI/ML and reinforcement-learning components.
10. Integration testing of the individual modules.
11. Mobile application testing, system testing, and performance evaluation.
12. Final documentation, result analysis, and project demonstration.

Chapter 6

RESULT AND CONCLUSION

6.1 Expected Outcome

The expected outcome of the proposed project is a mobile-based Financial Goal Planner that provides personalized, adaptive, and risk-aware investment planning for individual financial goals. The system is designed to consider the user's financial goal, target amount, investment horizon, savings capacity, and risk profile while generating suitable investment recommendations.

The proposed system is expected to integrate the following major functionalities:

- **Goal-Based Financial Planning:** The system will allow users to define financial goals by specifying the goal type, target amount, and expected time horizon.
- **Inflation-Adjusted Financial Calculation:** The system will consider the effect of inflation while estimating the future amount required to achieve the selected financial goal.
- **SIP and Savings Calculation:** The system will calculate the required periodic investment and savings contribution based on the target amount, investment horizon, and expected return.
- **Safety and Feasibility Assessment:** The system will evaluate whether the selected financial goal is practically achievable based on the user's available savings, required contribution, investment horizon, and other relevant financial parameters.
- **Risk Profiling:** The system will assess the user's risk tolerance and risk capacity using financial and goal-related information.
- **AI-Driven Recommendation:** The Recommendation Engine will generate personalized investment recommendations by considering the user's financial goal, risk profile, and investment horizon.
- **Dynamic Asset Allocation:** The system will adjust the recommended asset allocation according to the remaining goal horizon. As the goal deadline approaches, the allocation can be planned to become more conservative in order to reduce exposure to higher-risk assets.
- **Mobile Dashboard and Progress Tracking:** The mobile dashboard will provide users with information about their financial goals, required contributions, feasibility status, recommended allocation, and goal progress.

The Recommendation Engine will be inspired by the regret-optimized portfolio enhancement and future-looking reward concepts presented in the base paper. These concepts will be adapted to the proposed goal-based financial planning environment rather than directly replicating the original portfolio setting.

The expected system will therefore provide an integrated workflow in which financial calculations, feasibility assessment, risk profiling, investment recommendation, and dynamic allocation are connected through a single mobile-based platform.

6.2 Conclusion

Personal financial planning involves several factors such as financial goals, target amounts, investment horizons, expected returns, inflation, savings capacity, and risk tolerance. Conventional financial planning approaches may use fixed rules or static recommendations and may not adequately consider changes in the remaining goal horizon.

The literature reviewed in this project covers several important areas, including Artificial Intelligence, Reinforcement Learning, Deep Reinforcement Learning, portfolio optimization, robo-advisory, personal financial planning, investor profiling, risk-aware investment, and behavioural finance. These studies demonstrate the potential of intelligent techniques for improving financial decision-making and investment management.

However, the reviewed literature also indicates the need for an integrated approach that connects financial goal planning with feasibility assessment, risk profiling, personalized investment recommendation, and dynamic asset allocation. The proposed project addresses this requirement by designing a goal-oriented financial planning system that combines these components within a single mobile-based platform.

The proposed system is inspired by the base paper, “Regret-Optimized Portfolio Enhancement through Deep Reinforcement Learning and Future Looking Rewards”. The regret-based and future-looking reward concepts provide the foundation for designing an adaptive recommendation mechanism. In the proposed system, these concepts are considered in the context of individual financial goals, where the investment horizon and risk profile are important factors in determining the recommended allocation.

The major contribution of the proposed system is the integration of multiple financial planning components into one workflow. The system is planned to accept user-specific financial and goal-related inputs, perform the required calculations, assess feasibility, determine the risk profile, generate an investment recommendation, and dynamically adjust the allocation according to the remaining goal horizon.

Phase 1 of the project established the necessary foundation for the subsequent implementation. It included the study of the base paper, detailed literature review, identification of the research gap, formulation of the problem statement, definition of objectives, analysis of system requirements, design of the proposed architecture, identification of major modules, and preparation of the development workflow.

The subsequent phases will focus on the implementation and integration of the proposed modules. These phases will include mobile application development, financial calculation implementation, safety and feasibility assessment, risk profiling, recommendation-engine development, dynamic allocation, backend and database integration, model training and evaluation, system testing, and performance evaluation.

Thus, Phase 1 provides a clear research and technical foundation for developing the proposed Financial Goal Planner as an AI-driven, goal-based, and risk-aware investment planning system.

REFERENCES

- [1] A. Lamukanyani and Mantshimuli, "Sentiment-aware portfolio optimization: Cvar-based diversification with deep reinforcement learning," *IEEE Access*, 2025.
- [2] J. Ahn and H.-G. Kang, "Simul-rl portfolio framework: Black-scholes-merton and reinforcement learning for asset allocation," *IEEE Access*, 2025.
- [3] Y. Ansari, S. Gillani, M. Bukhari, B. Lee, M. Maqsood, and S. Rho, "A multifaceted approach to stock market trading using reinforcement learning," *IEEE Access*, 2024.
- [4] E. Benhamou, D. Saltiel, J.-J. Ohana, and J. Atif, "Detecting and adapting to crisis pattern with context based deep reinforcement learning," *arXiv.org*, 2020.
- [5] B. Dhashanamoorthi, H. Varghese, V. Venkateswaran, and S. Mandeep, "Ai-driven robo-advisors on personal financial management and investment strategies," *International Journal of Science and Research Archive*, 2024.
- [6] B. Enkhsaikhan and O. Jo, "Uncertainty-aware reinforcement learning for portfolio optimization," *IEEE Access*, 2024.
- [7] R. M. Gaspar and M. Oliveira, "Robo advising and investor profiling," *Multidisciplinary Digital Publishing Institute*, 2024.
- [8] M. C. Medeiros and G. Boaretto, "Forecasting inflation using disaggregates and machine learning," *arXiv.org*, 2023.
- [9] P. Yugopuspito and J. K. Jun, "Hybrid lstm and ppo networks for dynamic portfolio optimization," *arXiv.org*, 2025.
- [10] M. G. Alias Kanagaraj, S. Vasantha, and S. Raja, "Exploring the potential of ai in personal financial planning and advisory services," *arXiv.org*, 2024.
- [11] D. Karzanov, R. Garzón, M. Terekhov, C. Gulcehre, T. Raffinot, and M. Detyniecki, "Regret-optimized portfolio enhancement through deep reinforcement learning and future looking rewards," *arXiv.org*, 2025.
- [12] P. Kofman, "Scoring the ethics of ai robo-advice: Why we need gateways and ratings," *Journal of Business Ethics*, 2025.
- [13] M. S. Kulkarni, K. P. Patil, and D. Pramod, "The role of robo-advisors in behavioural finance and shaping investment decisions," *Cogent OA*, 2020.
- [14] E. Lwele, S. Emmanuel, and S. G. Sitali, "Risk-aware deep reinforcement learning for dynamic portfolio optimization," *arXiv.org*, 2025.
- [15] J. Olamijuwon and S. J. C. Zouo, "Machine learning in budget forecasting for corporate finance: A conceptual model for improving financial planning," *Open Access Research Journal*, 2024.
- [16] B. Patra, S. Sarkar, S. Pal, S. Ghosh, and S. Datta, "Ai-driven goal based financial planning system: A framework for contextual feasibility validation," *Rademics Research Institute*, 2025.

- [17] Y. Pereguda and M. Biloshkurskyi, “The impact of the digital economy on investment portfolio management in the stock market,” *Eastern European Union of Scientists*, 2024.
- [18] S. Samanta, S. Priya, N. Srivastava, S. K. Thakur, N. S. Simerpreet Kaur, S. Roy, and A. Kumar, “Effectiveness of sip investment among retail investors in jharkhand: A survey-based study,” *International Journal for Novel Research in Economics and Finance and Management*, 2024.
- [19] A. Sattar, A. Sarwar, S. Gillani, M. Bukhari, S. Rho, and M. Faseeh, “A novel rms-driven deep reinforcement learning for optimized portfolio management in stock trading,” *IEEE Access*, 2025.
- [20] J. P. Shetty, P. Singh, and S. Verma, “Robo-advisors in financial services: Redefining wealth management in the age of artificial intelligence,” *ScienceDirect*, 2026.
- [21] N. Solanki and M. Jha, “From time series to images: A gaf-resnet-lstm framework for asset selection and portfolio optimization,” *IEEE Access*, 2025.
- [22] F. Spadea and O. Seneviratne, “Parallel and multi-stage knowledge graph retrieval for behaviorally aligned financial asset recommendations,” *arXiv.org*, 2025.
- [23] P. Sunilkumar, “Ai revolution in wealth management: Applications and horizons,” *Journal of Information Systems Engineering and Management*, 2025.
- [24] M. Yin and M. Guo, “Complex forecasting and investment strategy optimization via chain-of-thought of large language models,” *ScienceDirect*, 2025.

APPENDIX A

PHASE 1 FLOW

The Phase 1 workflow of the group project can be summarized as:

1. Base paper study.
2. Literature review.
3. Research gap identification.
4. Problem statement formulation.
5. Objective identification.
6. System requirement identification.
7. Proposed architecture planning.
8. Financial calculation planning.
9. Safety and feasibility planning.
10. Risk profiling planning.
11. Recommendation engine planning.
12. Dynamic allocation planning.
13. Mobile application planning.
14. Phase 2 development planning.